

STUDENT NUMBER/NAME:

GOSFORD HIGH SCHOOL

2014

PRELIMINARY COURSE

FINAL EXAMINATION

Mathematics Extension 1

- **General Instructions**

- Reading time – 5 minutes
- Working time – 1½ hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- In Questions 6-9, show relevant mathematical reasoning and/or calculations

Total Marks – 53

Section I

5 marks

- Attempt Questions 1 – 5
- Allow about 10 minutes for this section

Section II

48 marks

- Attempt Questions 6 – 9
- Allow about 1 hour and 20 minutes for this section

Section I

5 marks

Attempt Questions 1 – 5.

Allow about 10 minutes for this section.

Use the separate multiple-choice answer sheet for Questions 1 – 5.

1. Which of the following is an expression for $\frac{9^n - 6^n}{9^n - 4^n}$?
- (A) $\frac{6^n}{4^n}$
- (B) $\frac{3^n}{5^n}$
- (C) $\frac{3^n}{3^n - 2^n}$
- (D) $\frac{3^n}{3^n + 2^n}$
2. If $R\cos(x + \alpha) \equiv \sin x - \cos x$, what are the values of $R\cos\alpha$ and $R\sin\alpha$?
- (A) $R\cos\alpha = -1$, $R\sin\alpha = -1$
- (B) $R\cos\alpha = -1$, $R\sin\alpha = 1$
- (C) $R\cos\alpha = 1$, $R\sin\alpha = -1$
- (D) $R\cos\alpha = 1$, $R\sin\alpha = 1$

3. What is the value of $\frac{{}^nC_3}{{}^nP_3}$?

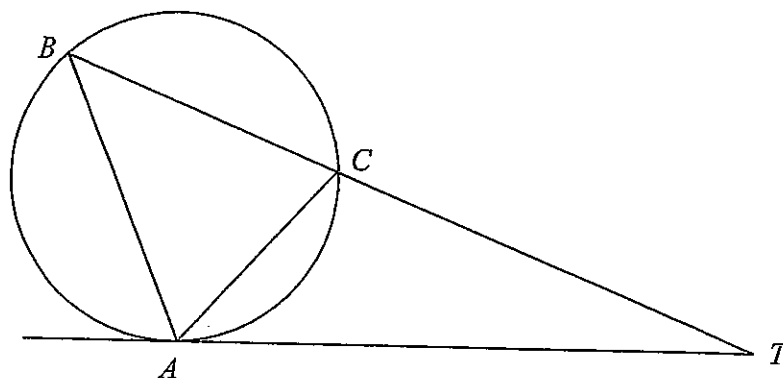
- (A) 6
- (B) 3
- (C) $\frac{1}{3}$
- (D) $\frac{1}{6}$

4. If $7\sqrt{x^5} - \sqrt{4x} + 3\sqrt{x^6} = a + b\sqrt{x}$, find a and b .

- (A) $a = 3x^3$, $b = 7x^2 - 4$
- (B) $a = 3x^3$, $b = 7x^2 - 2$
- (C) $a = 3x^2$, $b = 7x^2 - 2$
- (D) $a = 3x^6$, $b = 7x^4 - 4$

5.

NOT TO SCALE



In the diagram, ABC is a triangle inscribed in a circle. The tangent to the circle at A meets BC produced at T . $TA = 10$ cm and $TC = 8$ cm. What is the length of BC ?

- (A) 3.5 cm
- (B) 4 cm
- (C) 4.5 cm
- (D) 5 cm

END OF SECTION I

SECTION II

48 marks

Attempt Questions 6 – 9

Allow about 1 hour and 20 minutes for this section.

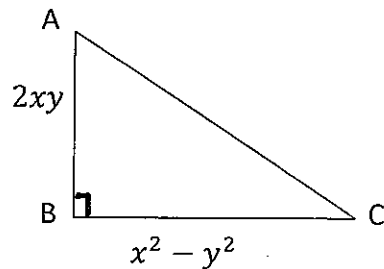
Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In questions 6 – 9, your responses should include relevant mathematical reasoning and/or calculations.

Question 6: (12 marks) (Use a SEPARATE writing booklet)

a. Solve the inequality $x - \frac{4}{x} \geq 0$ 2

b. Given the right angled triangle ABC 2



find an expression, in its simplest form, for AC.

c. The point A(4, 2) divides the interval XY internally in the ratio K : 1. If X is the point (10, -4) and Y is the point (0, 6), find the value of K, in its simplest form. 2

d. The angle between the lines
 $y = mx + 3$ and $x - 3y + 9 = 0$ is 45° . 2

Find the exact value(s) of m .

e. Simplify completely the expression
$$\frac{(n+1)! - n!}{(n-1)!}$$
 2

f. There are 8 people in a dinner group, 4 men and 4 women. In how many ways can they be seated around a circular table if a particular man Carl refuses to sit next to a certain woman Amanda. 2

END OF QUESTION 6

Question 7: (12 marks) (Use a SEPARATE writing booklet)

- a. If each distinct arrangement of the letters of

1

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is called a word, then how many words are possible?

- b. Find the number of ways in which the 7 letters of the word PYRAMID can be arranged in a row.

- i. without restriction

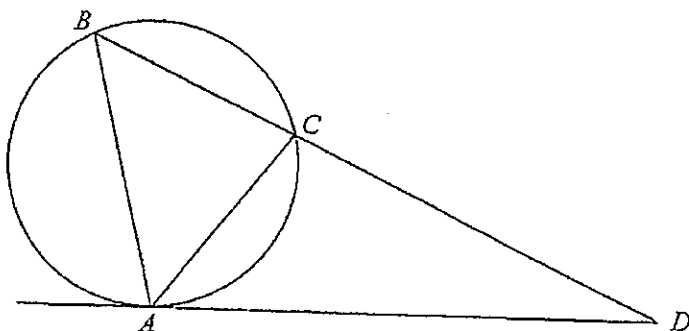
1

- ii. so that the 5 consonants are in alphabetical order from left to right.

1

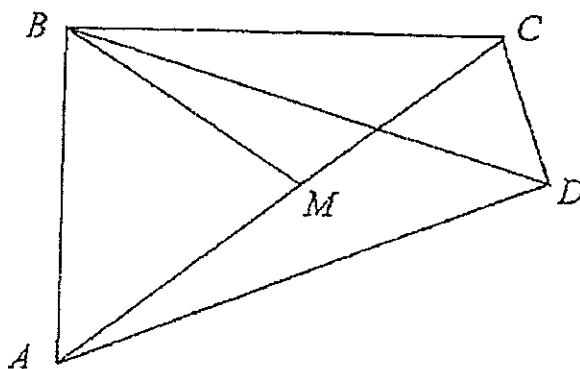
- c.

2



In the diagram, ABC is a triangle inscribed in a circle with $BC = AC$. The tangent to the circle at A meets BC produced at D . Show that AC bisects $\angle DAB$.

- d.



In the diagram, $ABCD$ is a quadrilateral in which $\angle ABC = \angle ADC = 90^\circ$. M is the midpoint of AC .

- i. Copy the diagram and explain why $ABCD$ is a cyclic quadrilateral.

1

- ii. Show that $\angle BMC = 2\angle BDC$

2

(Question 7 continues over the page)

Question 7 continued:

- e. i. Use the substitution $t = \tan \frac{x}{2}$
- to show that $\sec x + \tan x = \frac{1+t}{1-t}$ 2
- ii. Hence, solve the equation 2
- $\sec x + \tan x = -1$ for $0^\circ \leq x^\circ \leq 360^\circ$

END OF QUESTION 7

Question 8: (12 marks) (Use a SEPARATE writing booklet)

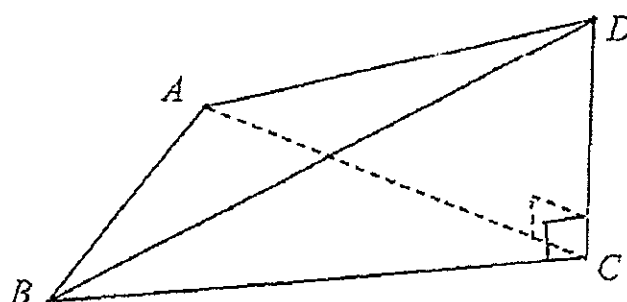
- a. Find the number of ways a committee can be chosen from 6 students, 5 parents and 4 teachers.

- i. without restriction. 1
- ii. so that the committee contains a majority of students. 2

- b. Solve the equation: 2

$$2\cos^2 x = -3 \sin x \text{ for } 0^\circ \leq x^\circ \leq 360^\circ$$

c.



In the diagram a vertical tower CD of height h metres stands with its bottom C on horizontal ground. A and B are two points on the ground such that $AB = 28$ metres.

From B the bearing of the tower is 050° and the angle of elevation of the top D of the tower is 30° . From A the bearing of the tower is 110° and the angle of elevation of the top D of the tower is 60° . Find the exact height of the tower.

- i. By first drawing a "bird's eye view" diagram indicating $\triangle ABC$ show that $\angle ACB = 60^\circ$. 1
- ii. Hence, find the exact height of the tower. 3

- d. i. Express $\tan(A + B)$ in terms of $\tan A$ and $\tan B$. 1
- ii. Hence, show that, if A , B and C are the vertex angles of a triangle then, 2
- $$\tan A + \tan B + \tan C = \tan A \tan B \tan C$$

END OF QUESTION 8

Question 9: (12 marks) (Use a SEPARATE writing booklet)

a. Factorise fully $54x^3 + 16$

1

b. i. Show that

$$\frac{1 + \tan x}{1 - \tan x} - \frac{1 - \tan x}{1 + \tan x} = 2 \tan 2x$$

2

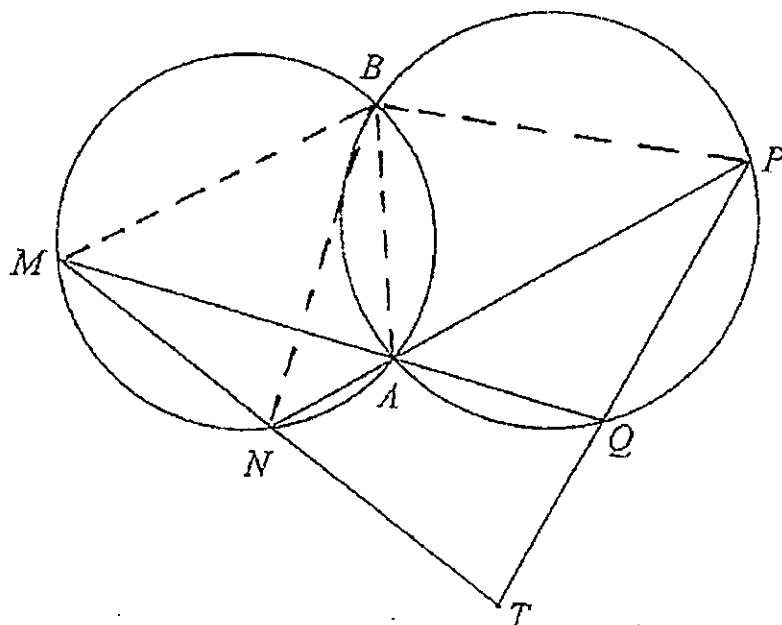
ii. Hence, find the value of:

$$\frac{1 + \tan 75^\circ}{1 - \tan 75^\circ} - \frac{1 - \tan 75^\circ}{1 + \tan 75^\circ}$$

1

in simplest exact form.

c.



3

In the diagram, two circles intersect at A and B. Points M and N lie on one circle and points P and Q lie on the other circle such that MAQ and NAP are straight lines. MN produced and PQ produced meet at T.

(Also, lines MB, NB, BA and BP are constructed for you.)

Show that TNBP is a cyclic quadrilateral.

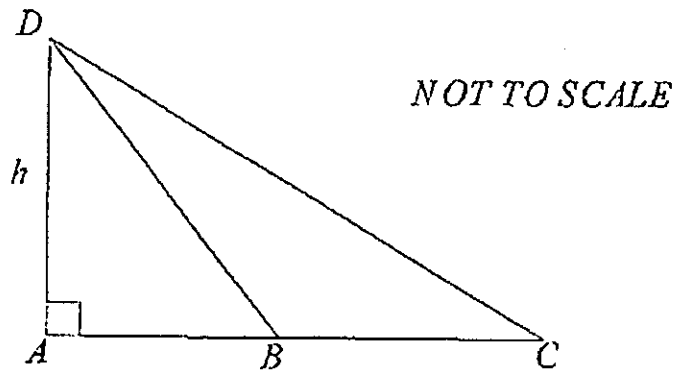
(Question 9 continues over the page)

Question 9 continued:

d.

(i) Prove that $\sin(45^\circ + \theta) \sin(45^\circ - \theta) = \frac{1}{2} \cos 2\theta$ 2

(ii) In the diagram, D is the top of a tower h metres high. The angles of depression of the points B and C are $(45^\circ + \theta)$ and $(45^\circ - \theta)$ respectively 3



Using the result in (i) show that $BC = 2h \tan 2\theta$

END OF EXAMINATION

Start here.

SECTION 1

$$\frac{9^n - 6^n}{9^n - 4^n} = \frac{3^n(3^n - 2^n)}{(3^n - 2^n)(3^n + 2^n)} = \frac{3^n}{3^n + 2^n} \quad (D)$$

$$\begin{aligned} 2. R \cos(x + \alpha) &= R(\cos x \cos \alpha - \sin x \sin \alpha) \\ &= R \cos x \cos \alpha - R \sin x \sin \alpha \end{aligned}$$

$$\begin{aligned} \therefore R \cos \alpha &= -1 \\ R \sin \alpha &= -1 \end{aligned} \quad (A)$$

$$\begin{aligned} 3. \frac{{}^nP_3}{{}^nC_3} &= \frac{\frac{n!}{(n-3)!3!}}{\frac{n!}{(n-3)!}} \\ &= \frac{1}{3!} \end{aligned}$$

$$= \frac{1}{6} \quad (D)$$

$$\begin{aligned} 4. 7\sqrt{x^5} - \sqrt{4x} + 3\sqrt{x^6} &= 7\sqrt{x^4 \cdot x} - 2\sqrt{x} + 3x^3 \\ &= 7x^2\sqrt{x} - 2\sqrt{x} + 3x^3 \\ &= 3x^3 + (7x^2 - 2)\sqrt{x} \end{aligned} \quad (B)$$

$$\therefore a = 3x^3 \quad b = 7x^2 - 2$$

$$5. AT^2 = BT \times CT$$

$$10^2 = BT \times 8$$

$$100 = 8 \times BT$$

$$12.5 = BT \quad (C)$$

$$\begin{aligned} \therefore BC &= 12.5 - 8 \\ &= 4.5 \text{ cm.} \end{aligned}$$

Answers.

1. D

2. A

3. D

4. B

5. C

SECTION 11

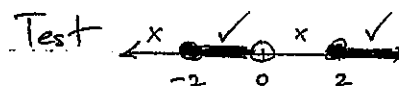
$$Q6. a) x - \frac{4}{x} \geq 0$$

$$\text{Consider } x - \frac{4}{x} = 0 \quad x \neq 0$$

$$x^2 - 4 = 0$$

$$x^2 = 4$$

$$x = \pm 2$$



$$-2 \leq x < 0 \text{ or } x \geq 2$$

b) Using Pythagoras

$$(2xy)^2 + (x^2 - y^2)^2 = AC^2$$

$$4x^2y^2 + x^4 - 2x^2y^2 + y^4 = AC^2$$

$$x^4 + 2x^2y^2 + y^4 = AC^2$$

$$(x^2 + y^2)^2 = AC^2$$

$$\therefore AC = x^2 + y^2$$

(positive since a length)

$$c) (10, -4) \quad (0, 6)$$

$$k:1$$

↓

$$(4, 2)$$

using the x co-ord:

$$1(10) + k(0) = 4$$

$$k+1$$

$$10 = 4k + 4$$

$$6 = 4k$$

$$\frac{6}{4} = k$$

$$\therefore k = \frac{3}{2}$$

$$(d) \text{ Using } \tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$\tan 45^\circ = \left| \frac{m - \frac{1}{3}}{1 + \frac{1}{3}m} \right|$$

$$1 = \left| \frac{3m - 1}{3 + m} \right|$$

$$1 = \left| \frac{3m - 1}{3 + m} \right|$$

$$\therefore \frac{3m - 1}{3 + m} = 1 \text{ or } \frac{3m - 1}{3 + m} = -1$$

$$3m - 1 = 3 + m \text{ or } 3m - 1 = -3 - m$$

$$2m = 4$$

$$4m = -2$$

$$m = 2$$

$$m = -\frac{1}{2}$$

$$\therefore m = 2 \text{ or } -\frac{1}{2}$$

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$$(e) \frac{(n+1)! - n!}{(n-1)!}$$

$$= \frac{n!(n+1-1)}{(n-1)!}$$

$$= \frac{n!(n)}{(n-1)!}$$

$$= n \times n$$

$$= n^2$$

(f) Total seatings = $7!$

Ways together = $6! \times 2$

$$\therefore \text{Ways apart} = 7! - 6! \times 2$$

$$= 3600$$

Q7 a) Arrangements = $\frac{7!}{3!2!}$

$$= 420$$

b) (i) $7! = 5040$

(ii) Ways the consonants can be arranged = $5!$

(alphabetical order is only 1 of these!)

You may ask for an extra Writing Booklet if you need more space.

Ways within the 7 letter arrangement

$$= \frac{7!}{5!}$$

$$= 42$$

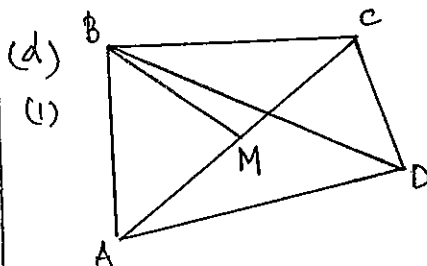
(c) let $\hat{CAD} = \theta$

$\therefore \hat{ABC} = \theta$ (alternate segment theorem)

also $\hat{BAC} = \theta$ (equal base $\angle$ s in isosceles $\triangle$)

$\therefore \hat{ABC} = \hat{BAC} = \theta$

$\therefore AC$ bisects $\hat{DAB}$



ABCD is a cyclic quadrilateral since $\hat{ABC} = \hat{ADC} = 90^\circ$ (given data)

and opposite angles in a cyclic quadrilateral are equal

Start here.

(ii) Since $\hat{ABC} = 90^\circ$

AC must be a diameter (angle in a semicircle = 90°)

$\therefore M$ is the centre of the circle (midpoint of diameter)

$$\therefore \hat{BMC} = 2 \times \hat{BDC}$$

(angle at the centre is twice the angle at the circumference standing on the same arc)

(e)(i) $\cos x = \frac{1-t^2}{1+t^2} \therefore \sec x = \frac{1+t^2}{1-t^2}$

$$\tan x = \frac{2t}{1-t^2}$$

$$\therefore \sec x + \tan x = \frac{1+t^2}{1-t^2} + \frac{2t}{1-t^2}$$

$$= \frac{1+2t+t^2}{1-t^2}$$

$$= \frac{(1+t)^2}{(1-t)(1+t)}$$

$$= \frac{1+t}{1-t}$$

as req.

(ii.) $\therefore \frac{1+t}{1-t} = -1$

$$1+t = -1+t$$

$$0 = -2$$

$\therefore$ no solutions

but we must test $x = 180^\circ$

LHS =

$$\sec 180^\circ + \tan 180^\circ$$

$$= -1 + 0$$

$$= -1$$

$\therefore 180^\circ$ is a solution

(the only solution)

Q8

(a) (i) ${}^{15}C_3 = 455$

(ii) there can be 3 students or 2.

$$\therefore {}^6C_3 + {}^6C_2 \times {}^9C_1$$

$$= 20 + 15 \times 9$$

$$= 155$$

(b) $2\cos^2 x = -3\sin x$

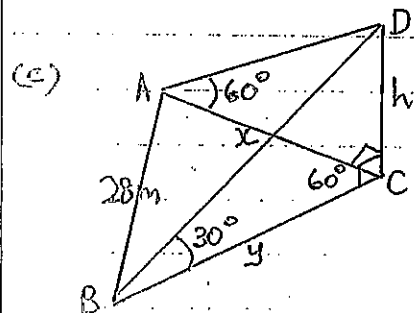
$$2(1-\sin^2 x) = -3\sin x$$

$$0 = 2\sin^2 x - 3\sin x - 2$$

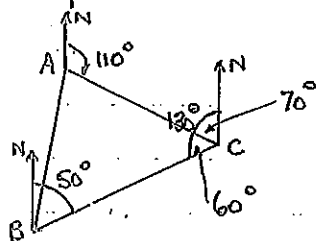
$$0 = (2\sin x + 1)(\sin x - 2)$$

$$\therefore \sin x = -\frac{1}{2} \text{ or } \sin x = -2 \text{ (no solns)}$$

$$\therefore x = 210^\circ, 330^\circ$$



This diagram would also help.



From the top diagram

$$\tan 60^\circ = \frac{h}{x}$$

$$x = \frac{h}{\tan 30^\circ}$$

$$x = \frac{h}{\sqrt{3}}$$

$$\text{Similarly } y = \sqrt{3}h$$

Using the cosine rule

$$28^2 = x^2 + y^2 - 2xy \cos 60^\circ$$

$$784 = \frac{h^2}{3} + 3h^2 - 2h^2 \times \frac{1}{2}$$

$$784 = \frac{10h^2}{3} - h^2$$

$$784 = \frac{7h^2}{3}$$

$$2352 = 7h^2$$

$$336 = h^2$$

$$\sqrt{336} = h$$

$$\therefore h = 4\sqrt{21} \text{ m}$$

$$(d) (i) \tan(A+B)$$

$$= \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$1 - \tan A \tan B$$

$$(ii) \text{ if } A, B, C \text{ are}$$

the 3 angles of a Δ

then $A+B+C = 180^\circ$

$$\therefore A+B = 180^\circ - C$$

$$\therefore \frac{\tan A + \tan B}{1 - \tan A \tan B} = \tan(180^\circ - C)$$

$$\therefore \frac{\tan A + \tan B}{1 - \tan A \tan B} = -\tan C$$

$$\tan A + \tan B = -\tan C (1 - \tan A \tan B)$$

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$$\therefore \tan A + \tan B = -\tan C + \tan A \tan B \tan C$$

$$\therefore \tan A + \tan B + \tan C = \tan A \tan B \tan C \quad (c) \hat{MAB} = \hat{MNB}$$

as req.

($\angle$'s at the circumference standing on the same arc are equal)

Q9

$$(a) 54x^3 + 16$$

$$= 2(27x^3 + 8)$$

$$= 2(3x+2)(9x^2 - 6x + 4)$$

$$(b) LHS = \frac{(1+\tan x)^2 - (1-\tan x)^2}{(1-\tan x)(1+\tan x)}$$

$$= \frac{1+2\tan x + \tan^2 x - 1+2\tan x - \tan^2 x}{(1-\tan x)(1+\tan x)}$$

$$= \frac{4\tan x}{1-\tan^2 x}$$

$$= 2 \times \frac{2\tan x}{1-\tan^2 x}$$

$$= 2 \times \tan 2x$$

$$= 2 \tan 2x \text{ as req.}$$

$$(ii) \therefore x = 75^\circ$$

$$LHS = 2 \tan 150^\circ$$

$$= 2 \times -\frac{1}{\sqrt{3}}$$

$$= -\frac{2}{\sqrt{3}}$$

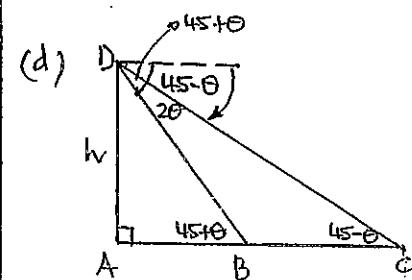
$$= -\frac{2\sqrt{3}}{3}$$

$\hat{MAB} = \hat{BPQ}$ (exterior $\angle$ of cyclic quadrilateral is equal to interior opposite $\angle$)

and $\therefore \hat{MNB} = \hat{BPQ}$

and $\therefore$ TNBP is a cyclic quadrilateral

Since exterior $\angle$ of cyclic quadrilateral = interior opposite angle.



$$(1) \sin(45+\theta) \sin(45-\theta) = (\sin 45^\circ \cos \theta + \sin \theta \cos 45^\circ) \times (\sin 45^\circ \cos \theta - \sin \theta \cos 45^\circ)$$

$$= \sin^2 45^\circ \cos^2 \theta - \sin^2 \theta \cos^2 45^\circ$$

$$= \frac{1}{2} \cdot \cos^2 \theta - \frac{1}{2} \sin^2 \theta$$

You may ask for an extra Writing Booklet if you need more space.

Start here.

$$= \frac{1}{2} (\cos^2 \theta - \sin^2 \theta)$$

$$= \frac{1}{2} \cos 2\theta \text{ as req.}$$

(ii) In $\triangle ADB$

$$\sin(45+\theta) = \frac{h}{DB} \quad \text{--- (1)}$$

In $\triangle BDC$, using sine rule

$$\frac{BC}{\sin 2\theta} = \frac{DB}{\sin(45-\theta)} \quad \text{--- (2)}$$

From (1)

$$DB = \frac{h}{\sin(45+\theta)}$$

$$\text{In (2)} \quad \frac{BC}{\sin 2\theta} = \frac{h}{\sin(45+\theta) \sin(45-\theta)}$$

$$\frac{BC}{\sin 2\theta} = \frac{h}{\frac{1}{2} \cos 2\theta}$$

$$BC = \frac{2h \sin 2\theta}{\cos 2\theta}$$

$$= 2h \tan 2\theta$$

as req.